

Integrative Model: The Human-Human DigiPPPiP Kernel

Integrative Model: The Human-Human DigiPPPiP Kernel I

DigiPPPiP can be summarized as a cyberphysical relational practice. Its first layer is a coupled modeling lens: partners can be described as updating expectations about each other's intentions through shared marks, with the formal primitive isolated in [sec:formalisms_appendix]. Its second layer is cyber-phenomenological co-presence: marks carry embodied gesture across physical, digital, and hybrid surfaces. Its third layer is narrative information architecture: sessions become symbolic trajectories with entropy, surprise, and convergence.

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The fourth layer is neuroergonomic design: the interface should preserve attention, reduce technoferece, and support flow. The fifth layer is place-relational grounding: sessions can bind two bodies, two environments, and one artifact. The sixth layer is accessibility: input and feedback should adapt to participants rather than forcing participants to adapt to one canonical interface. The word “accessibility” is used here as a design and validation program, not as a universal guarantee.

The seventh layer is temporal flexibility. Synchronous, semisynchronous, and asynchronous modes are not rank-ordered; they answer different relational constraints. The eighth layer is measurement. Active-inference, narrative-information, event-log, accessibility, outcome-model, and geometric-hyperscanning tools give the framework empirical handles without letting the manuscript pretend that conceptual simulations are evidence.

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The model can also be collapsed to a minimum viable DigiPPPiP loop: partner acts, mark persists, other partner perceives, response changes the shared field, and both partners retain agency over what remains. Every larger layer should be justified by how it strengthens that loop. If a feature does not improve perception, action, persistence, interpretation, access, or consent, it is outside the core framework even if it is technically impressive. The architecture in [fig:cpss_architecture] operationalizes that rule by separating the kernel from instrumentation, modeling, optional AI support, and publication governance.

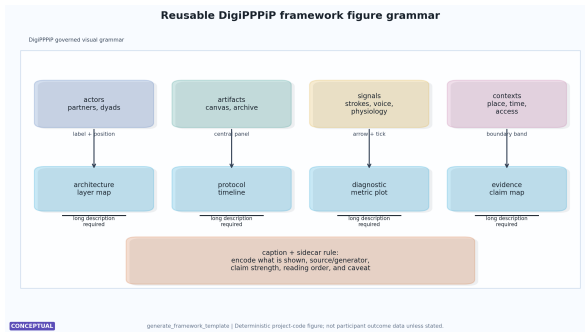
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The systems-governance layer makes that collapse enforceable. `src/systems_governance.py` treats the human-human mark loop as the kernel and places event logs, place context, optional physiology, AI assistance, and clinical translation in named boundary positions. This matters because a system boundary is a design decision, not a discovered fact: moving physiology, AI, or clinical translation into the kernel would change the participant contract and the evidence burden. The integrative model therefore remains reversible. Branches can be removed, disabled, coarsened, or downgraded without losing the core practice.

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[@fig:framework_template] summarizes the reusable visual grammar that holds these layers together. Actors, artifacts, signals, contexts, diagnostics, and evidence maps are styled consistently so captions can identify what is encoded, which generator produced it, what claim strength it supports, and what caveat remains. The grammar now has a stronger architecture role: human actions, optional AI branches, computed diagnostics, and publication artifacts should be visually separable even when they appear in one system figure.

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Figure 1: Reusable DigiPPPiP figure grammar for actors, artifacts, signals, contexts, architecture maps, protocol timelines, diagnostic plots, and evidence maps. The figure is generated by `generate_framework_template()`; read the upper row as semantic roles, the small labels as non-color channels, the middle row as reusable figure forms, and the lower box as the caption and long-description rule applied throughout the manuscript. It supports integrative consistency, not empirical validation, and future data figures should keep the same claim-boundary discipline.